

Rotman

CODING WITH AI – A QUICK CHAT

“..., but you cannot outsource your understanding”

September 29, 2026 Prepared by Jay Cao / [MDAL](#)

Website: <https://rmdal.github.io/mma-dv-2026/>



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Rapid Evolution: Three Levels

Level	Primary Capability	Typical Examples	Human Role
Inline Completion	Suggests next lines or snippets in real time	Autocomplete, copilot-style suggestions	Full control over design, logic, and review
Assisted Coding	Generates and explains larger code blocks on request	Function generation, refactoring, test creation	Guides prompts, integrates code, validates results
Agentic Coding	Plans, writes, tests, and iterates autonomously	Multi-file changes, automated debugging, task execution	Defines goals, constraints, and provides oversight

Rapid Progress

“I started leaning hard into Claudifying about a year ago. That’s been a crazy adventure and it’s now been ~5 months since I last wrote any code myself.”

Anthropic employee*

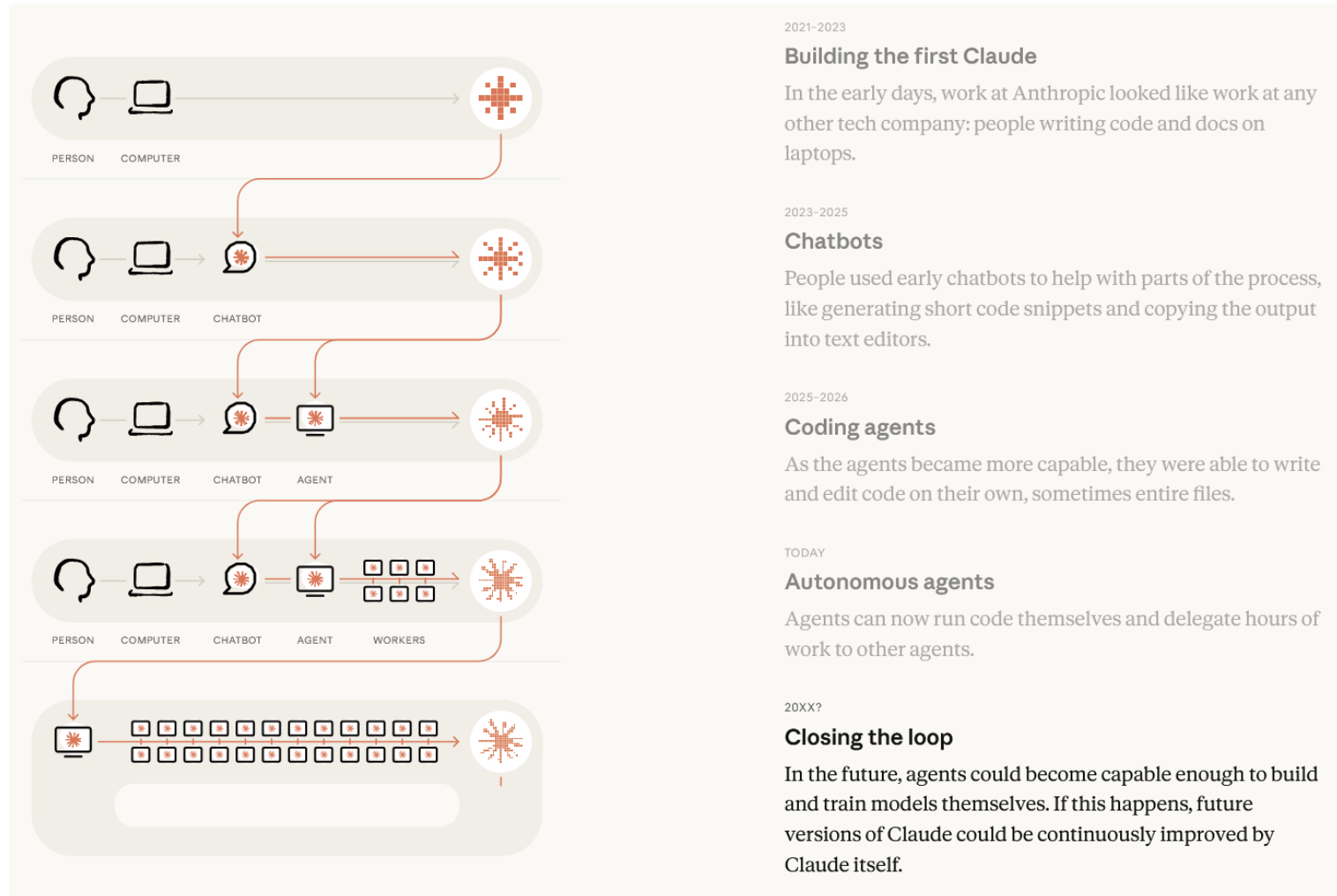
“Claude-written code was somewhat worse than human-written code at Anthropic in late 2025, is roughly at parity today, and we expect it to be strictly better within the year.”

“The shape of stuff today is roughly ‘humans have ideas, and the models are able to implement, test and evaluate them an [order of magnitude] faster than before.’”

“The comparative advantage of humans as of right now is still in seeing the bigger picture and thinking beyond the confines of the immediate task.”

Source: <https://www.anthropic.com/institute/recursive-self-improvement> (June 2026)

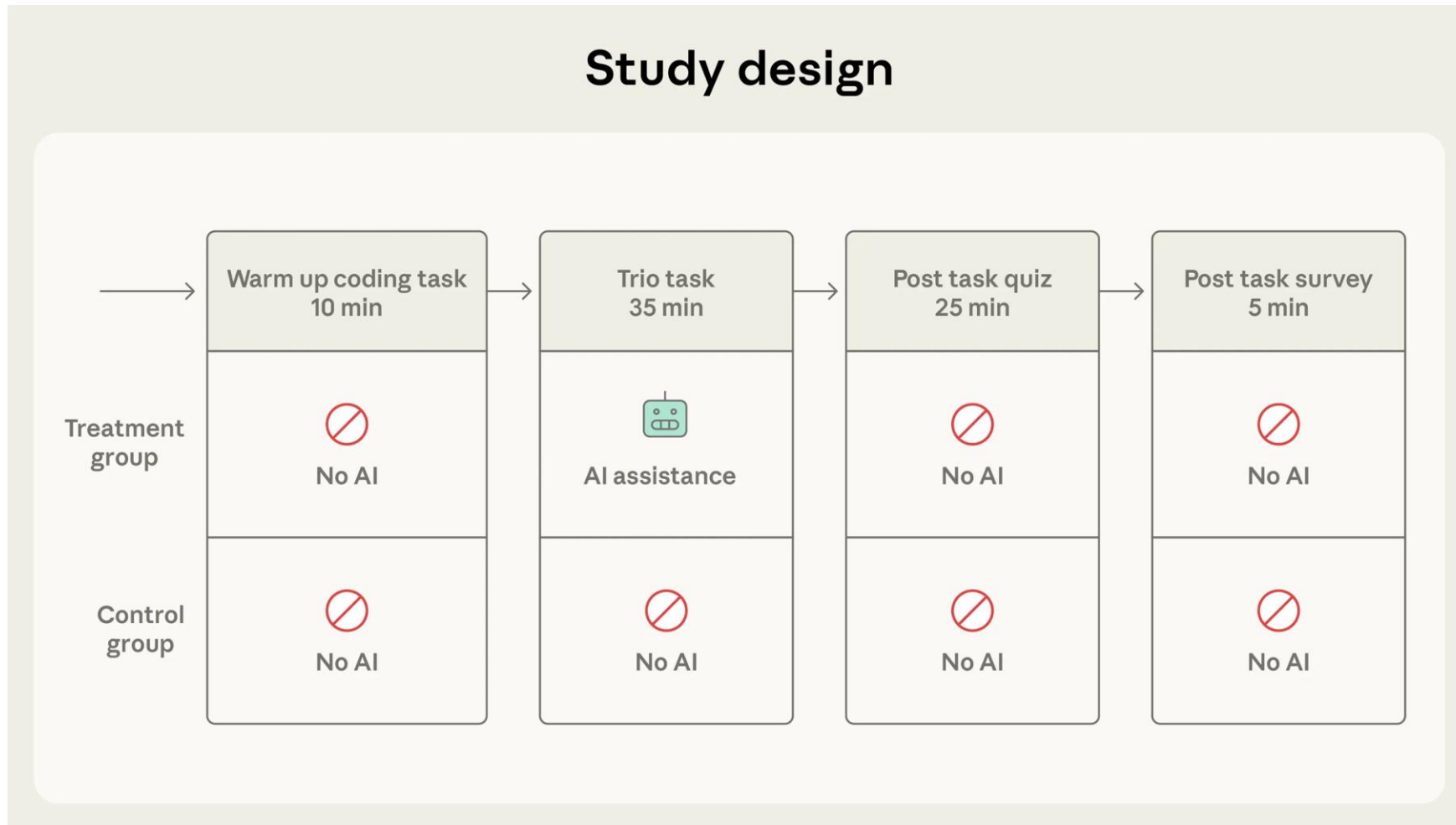
Distant or Near Future?



How do you be relevant here?

Source: <https://www.anthropic.com/institute/recursive-self-improvement> (June 2026)

AI Use -> Coding Speed & Skill Formation - 1

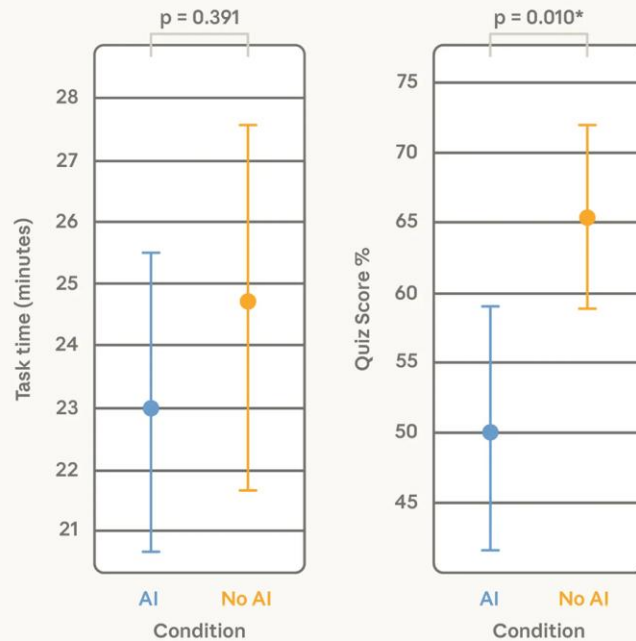


Source: <https://www.anthropic.com/research/AI-assistance-coding-skills> (Jan 2026)

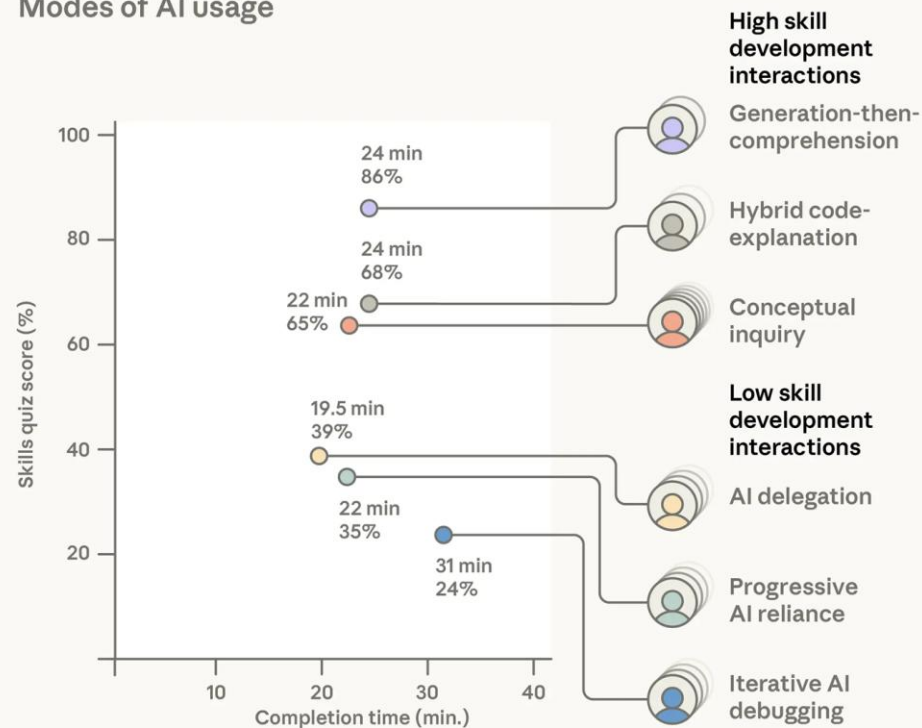
AI Use -> Coding Speed & Skill Formation - 2

How AI assistance impacts coding speed and skill formation

AI assistance: treatment effect on coding speed and knowledge score



Modes of AI usage



You want to be here at least, for now.

Why

- Skill formation emerges from the struggle to resolve a problem
 - Skill forms as learners attempt, fail, and revise their solutions
- Skill formation does not come from passive exposure to its solution.
 - being handed the correct answer cuts the “productive struggle” process short and leads to weaker learning

How Do You Plan to Use AI (for Coding & ...)?

- Perhaps we all know the right answer
 - Do
 - Generation then comprehension (by asking follow-up questions of the code)
 - Conceptual inquiry
 - Hybrid code explanation (i.e., code generation along with explanations)
 - Don't
 - AI delegation
 - Progressive AI reliance
 - AI debugging without clarifying one's own understanding
- Reality could be harsh
 - So much workload (assignments, presentation...)
 - Everyone is using it
 - ...

Renter vs Builder – Who Do You Want to Be?

- Renter
 - They copy and paste AI-generated answers directly. They save time initially but rent a brain that disappears the moment the WiFi drops or someone probes their logic
- Builder
 - They use AI to accelerate thinking. They review the output, verify the facts, and grapple with the logic to learn further insights. This process compounds over time, building lasting skill.

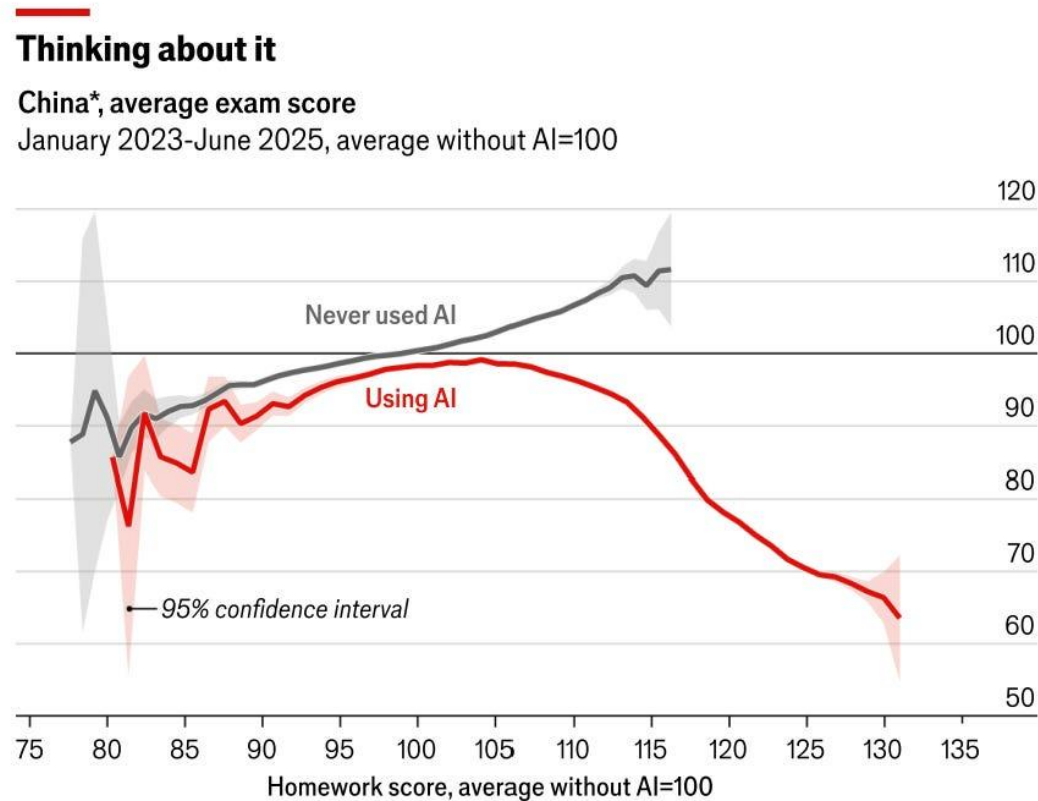
Don't Offload Your Brain to AI – Be Relevant

- “Think outside the bots ”
 - “Mental friction” is required for true creativity, memory, and critical thinking
 - Using AI as a tool to pressure-test your own ideas
- “You can outsource you thinking, but you cannot outsource your understanding”
 - -> fast, cheap outputs
 - -> integrate, test, and defend the ideas yourself



Source: ['Think outside the bots': How to stop AI from turning your brain to mush](#)

Appendix – Another Recent Study in the News



*26,811 students aged 12-18

Source: "The generative AI learning penalty: evidence from Chinese secondary education",
by D. Strömberg et al., preprint June 2026

Source: [Does AI stop children from learning?](#)

An essay by Benjamin Riley: [The 50 Percent Problem](#)

Appendix – Recent in the News

Goldman Sachs partner warned that AI is putting bankers' reasoning skills at risk

Chris Churchman, who leads Goldman's Marquee platform, said AI risks causing "cognitive atrophy" among junior bankers who outsource thinking to algorithms

By Cris Tolomia · 2 min read · Updated August 24, 2026

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Source: <https://qz.com/goldman-sachs-ai-reasoning-skills-bankers-082426>

Appendix – AI for Math: Good or Bad

← Continued thread



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🌐 1d

As Hugo Duminil-Copin wrote recently at proofsandprompts.com/2026/08/3..., the indiscriminate automated strip-mining of open problems for solutions may destroy the ecosystem from which the next generation of mathematical techniques, problems, and practitioners would have developed, similarly to how using excavators to dig out treasures from an archeological site destroys the rich historical context and data that gives such treasures so much of their meaning and value.

It may become necessary to declare certain classes of mathematical problems off-limits to automated solvers, in order to preserve their broader value to the mathematical ecosystem (for instance, through the training of future mathematicians). Admittedly this can be hard to enforce when such tools are both powerful and widely available. But then again.

Source: <https://mathstodon.xyz/@tao/117204929023813310>;

Good read: [A Severe Misalignment of AI in Mathematics](#); [here](#) and [here](#);